ZDC autoresearch, Phase 6: the crystal-seeded ZDC photon finder

Find the ZDC photons the hadronic section loses, with the LYSO crystals in front of it

Started 2026-11-09 · one slide per try · rebuilt from the run's result files

Phase 6 tries — Λ found on the search halves, step by step

try	change	9x275	9x130	score	purity	z	−t σ68 9x275	verdict
Z1	crystal-seeded photons in the ZDC Λ pairing	1267 → 1271	219 → 227	1486 → 1498	26.7 → 26.6 %	+0.8	0.124 GeV²	INCONCLUSIVE
Z1b	crystal photons with the Hcal column from the rec hits	1267 → 1314	219 → 232	1486 → 1546	26.7 → 27.5 %	+3.6	0.126 GeV²	ACCEPTED
Z2	crystal photon parameters: threshold, column width, neutron share	1314 → 1388	232 → 230	1546 → 1618	27.5 → 28.0 %	+5.6	0.122 GeV²	ACCEPTED
Z4	the ZDC pairing's Λ-mass width 0.04 → 0.03 GeV	1388 → 1409	230 → 237	1618 → 1646	28.0 → 27.1 %	+3.2	0.122 GeV²	ACCEPTED
Z5	the crystal law's energy as a competing hypothesis for Hcal photons	1409 → 1574	237 → 252	1646 → 1826	27.1 → 29.5 %	+11.6	0.147 GeV²	ACCEPTED
Z6	the crystal-law photon's Hcal column leaves the neutron	1574 → 1547	252 → 250	1826 → 1797	29.5 → 29.1 %	-2.8	0.121 GeV²	REJECTED

Score 1486 → 1826 (+23 %) since the start row Y9; −t σ68 at 9x275 0.124 → 0.147 GeV² (Q22).

Phase 6 start row Y9 — yield, purity and −t resolution per beam

Y9 on the search halves D6s	9x275	9x130
N_found / N_expected	1267 / 7160	219 / 7160
yield (reachable)	17.70 % (43.6 %)	3.06 % (8.0 %)
purity in the ±30 MeV window	26.4 %	28.7 %
−t resolution σ68 of found Λ	0.12 GeV²	0.13 GeV²
−t within ±0.1 GeV²	56 %	58 %

Kept from Phase 5

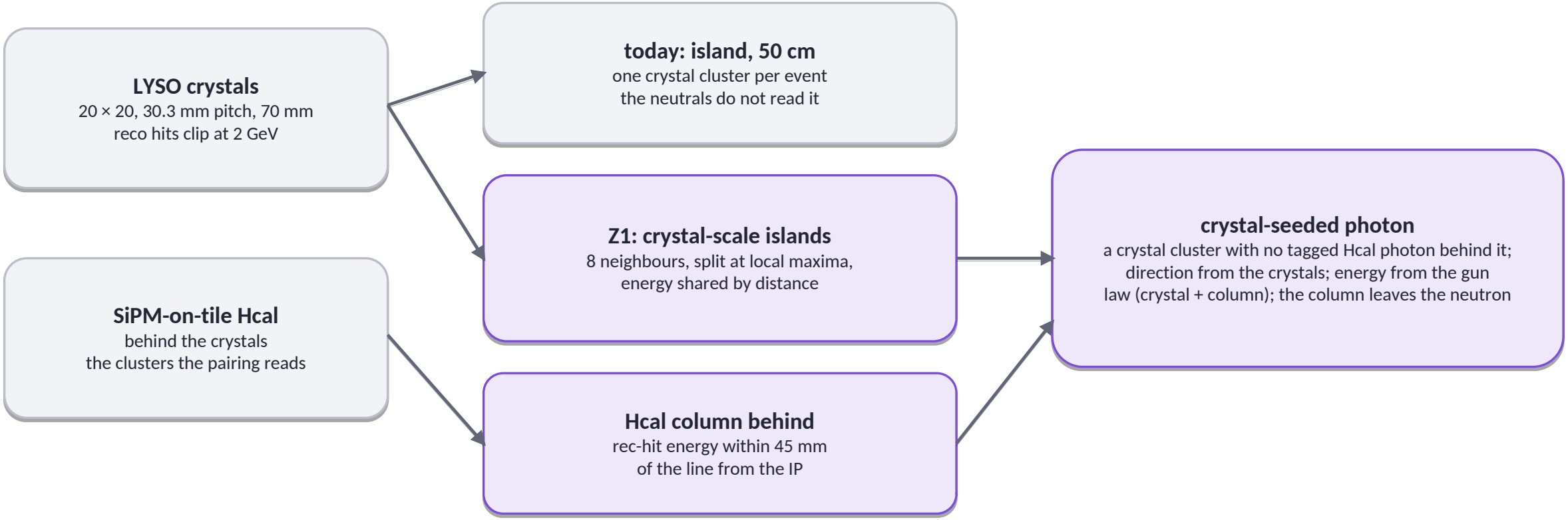
- score: N_found pooled over D6s 9x275 + 9x130
- selector 463: paired z ≥ 3, purity z_P ≥ −2
- held out: D6h, checkpoint every 5 tries
- fair play: 460 F1–F6, 465 patch scan
- new: 468, F3 for crystal clusters

−t resolution at 9x275 by true −t (GeV², σ68)

- 0.0–0.2: 0.05, 0.2–0.5: 0.08, 0.5–1.0: 0.11, 1.0–3.0: 0.18
- beam spread 0.045 GeV², detector 0.110 GeV² (476)

The all-ZDC regime holds 28 % of expected at 9x275; a ZDC photon without a photon neutral is its largest loss.

LYSO crystal section — today's clustering and a crystal-seeded photon



In the pairing

- new hypotheses beside Y9's: crystal + tagged Hcal photon, crystal + crystal, crystal + B0 photon; the lowest score wins as before

Step 0 scope study — crystal candidates for the photons Y9 misses (9x275)

ZDC photons of a Λ → nπ⁰ (search half)	all in the ZDC	one γ in B0
without a matched photon neutral in Y9	1422	265
… with a pure crystal cluster ≥ 0.5 GeV	637	167
Λ missed with both photons available (oracle)	708	270

Emulated gain, best variant (veto split crys_pd -10)

- all-ZDC +116, one γ in B0 +7, both γ in B0 -3: net +120 found Λ
- emulator check: Y9's own decision gives 750 / 395 against the row's 749 / 395
- stop rule: below +50 pooled the phase stops here

Z1 — crystal-seeded photons in the ZDC Λ pairing

Change

- crystal section clustered at the crystal scale: 8 neighbours, split at local maxima ≥ 0.1 GeV
- a crystal cluster ≥ 0.5 GeV with no tagged Hcal photon behind it is a photon candidate
- its energy 1.962·E_crys + 47.47·E_column − 0.881 (gun fit, D2 + D2c); its column leaves the neutron
- new hypotheses: crystal + Hcal photon, crystal + crystal, crystal + B0 photon

Prediction (written before launch)

- 477 emulation: N_found 1267 → ~1385, 219 → ~234 (+133 pooled); purity 26.7 → ~28.4 %

search half	best	this try
9x275 N_found	1267	1271
9x275 purity	26.4 %	26.2 %
9x130 N_found	219	227
9x130 purity	28.7 %	29.0 %

paired: b 110, c 122, z +0.79; purity z_P -0.17

INCONCLUSIVE

fair play PASS

−t σ68 at 9x275: 0.124 GeV² (476)

Z1b — crystal photons with the Hcal column from the rec hits

Change

- Z1 summed the Hcal column over the clusters' HEXPLIT subcells: 0.45 of the rec-hit column
- its crystal photons carried a median 0.56 of their energy (Z1: +12, INCONCLUSIVE)
- Z1b sums the column over HcalFarForwardZDCRecHits (6th input), the hits the law was fitted on

Prediction (written before launch)

- as Z1's emulation: N_found 1267 → ~1385, 219 → ~234 (+133 pooled); purity 26.7 → ~28.4 %

search half	best	this try
9x275 N_found	1267	1314
9x275 purity	26.4 %	27.1 %
9x130 N_found	219	232
9x130 purity	28.7 %	29.8 %

paired: b 109, c 169, z +3.60; purity z_P +0.91

ACCEPTED | **fair play PASS**

−t σ68 at 9x275: 0.126 GeV² (476)

Z2 — crystal photon parameters: threshold, column width, neutron share

Change

- crystal candidates from 0.3 GeV (0.5 GeV)
- Hcal column 60 mm wide (45 mm), the law refitted on the guns at that width
- the column leaves the neutron on the Hcal clusters' scale (× 1/0.0273): no double counting

Prediction (written before launch)

- 477: +31 emulated over Z1b; at Z1b's 45 % fidelity ~+14: 1314 → ~1328, 232 → ~232; purity flat

search half	best	this try
9x275 N_found	1314	1388
9x275 purity	27.1 %	27.8 %
9x130 N_found	232	230
9x130 purity	29.8 %	28.7 %

paired: b 48, c 120, z +5.55; purity z_P +0.55

ACCEPTED | **fair play PASS**

−t σ68 at 9x275: 0.122 GeV² (476)

Z4 — the ZDC pairing's Λ-mass width 0.04 → 0.03 GeV

Change

- with crystal candidates the chooser weighs more hypotheses per event
- 679 missed Λ at 9x275 hold pure candidates for both photons
- a narrower Λ-mass term ranks the true triplet first more often

Prediction (written before launch)

- 477 --full: +27 found, +308 in the window: 1388 → ~1409, 230 → ~236; purity 28.0 → ~27 %

search half	best	this try
9x275 N_found	1388	1409
9x275 purity	27.8 %	26.9 %
9x130 N_found	230	237
9x130 purity	28.7 %	27.9 %

paired: b 24, c 52, z +3.21; purity z_P -1.11

ACCEPTED **fair play PASS**

−t σ68 at 9x275: 0.122 GeV² (476)

Z5 — the crystal law's energy as a competing hypothesis for Hcal photons

Change

- a tagged Hcal photon with a crystal cluster in front also competes with the crystal law's energy
- crystal law σ 10–11 % on the guns at 5–50 GeV; the Hcal law 24–34 %
- the chooser keeps whichever energy fits; the winner links the crystal cluster too
- Z3, the outright replacement, lost 107 Λ at 9x275 in emulation

Prediction (written before launch)

- 477 --alt: +160 over Z4 at a nearly flat window: 1409 → ~1556, 237 → ~250; purity up

search half	best	this try
9x275 N_found	1409	1574
9x275 purity	26.9 %	29.5 %
9x130 N_found	237	252
9x130 purity	27.9 %	29.7 %

paired: b 30, c 210, z +11.62; purity z_P +3.05

ACCEPTED | **fair play PASS**

−t σ68 at 9x275: 0.147 GeV² (476)

Z6 — the crystal-law photon's Hcal column leaves the neutron

Change

- Z5 measures the Λ energy high (reco / true 1.110 vs Z4's 1.073) and −t worse at 9x275
- a crystal-law Hcal photon's column beyond its own cluster stayed in the neutron: counted twice
- Z6 removes it from the neutron, as Z2 does for crystal photons

Prediction (written before launch)

- 477: Λ energy 1.122 → 1.058 for −14 found at 9x275: no gain for 463; −t below Z4's 0.122 GeV²

search half	best	this try
9x275 N_found	1574	1547
9x275 purity	29.5 %	29.0 %
9x130 N_found	252	250
9x130 purity	29.7 %	29.1 %

paired: b 68, c 39, z -2.80; purity z_P -0.58

REJECTED

fair play PASS

−t σ68 at 9x275: 0.121 GeV² (476)

Λ count against −t resolution — Y9, Z5 and Z6 on the search halves

row	N_found (9x275 + 9x130)	purity	−t σ68 9x275	−t σ68 9x130	−t within ±0.1 GeV², 9x275
Y9 (start)	1486	26.7 %	0.124 GeV²	0.133 GeV²	56 %
Z5 (current best)	1826	29.5 %	0.147 GeV²	0.128 GeV²	51 %
Z6 (REJECTED by 463)	1797	29.1 %	0.121 GeV²	0.120 GeV²	60 %

Why Z5 measures −t worse and Z6 does not

- Z5 gives tagged Hcal photons the crystal law's energy when it fits better; their Hcal column beyond the cluster stays in the neutron
- the Λ energy reads high: reco / true median 1.110 (Z4 1.073, same Λ); Z6 removes that column from the neutron
- Z6 loses 29 found Λ against Z5 (z −2.80) and measures −t better than the start row

Q22 for the user: the base for the next tries — Z5 (more Λ) or Z6 (better −t).

Checkpoint 1 — Z5 on the held-out MSF halves and the D4 gun

N_found (yield)	Y9, search	Z5, search	Y9, held out	Z5, held out	reachable
9x275	1267 (17.70 %)	1574 (21.98 %)	1206 (16.84 %)	1520 (21.23 %)	43.6 %
9x130	219 (3.06 %)	252 (3.52 %)	205 (2.86 %)	235 (3.28 %)	8.0 %
purity, held out (pooled)			25.8 %	28.8 %	

held out: REPRODUCES

b 83, c 427, z +15.23; purity z_P +3.59

D4 Λ gun, S4a_tm per thrown Λ → nπ⁰ (reported, never selected on; 410 credits Hcal clusters only)

- Y9: 7.44 % mean, 14.12 % at 275 GeV; Z5: 7.71 % mean, 14.56 % at 275 GeV

The held-out halves never chose anything; they confirm or revert the gain of the search halves.